SYSEN 5888 Deep Learning Group Project Plan

Model Evaluation and Improvements of Piczak (2015), "Environmental Classification with CNNs"

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Introduction

Our team will apply modern deep learning models and algorithms to demonstrate improvements in deep learning's ability to recognize and categorize sounds in audio files. We will first replicate the results of a 2015 study by Dr. Karol Piczak of Warsaw University of Technology titled "ESC: Dataset for Environmental Sound Classification" [1]. At the time its publishing, research into environmental sounds was lagging behind speech and music processing, in large part due to a lack of clean, labeled data. Dr. Piczak's paper introduced the ESC-50 dataset, a cleaned dataset comprising 2000 annotated clips comprising 50 classes of common sounds paired with 250,000 unlabeled sound files. This dataset was used to train a 2-layer CNN and compare performance to a human. After replicating this result, our group will test several model architectures against the original CNN baseline to evaluate our hypothesis that advancements in neural network technologies over the past 10 years will yield significant performance improvements.

Problem Statement and Background

For this effort, we intend to show that AI models can be trained to correctly recognize and classify sounds commonly heard in daily environments more accurately given advances in AI models and processing power. The models will be provided with short audio clips and will predict the type of sound in each file. After listening to the short clip, the model will attempt to classify what it heard into one of fifty different classes, such as a barking dog, rain, emergency vehicles, keyboard typing, etc. And while this would appear to be a straightforward task, it will be made more complicated for the model due to the background noise. Not only will the model need to correctly identify the sound classification in the file, but it will also have to do so while filtering out irrelevant background noise.

This is a problem worth investigating as there are numerous practical applications for this type of technology. Systems that can correctly identify sounds in their environments could be particularly useful in instances safety and monitoring or when machines and people interact near one another, systems such as robotics or assistive technologies. It could be helpful to the hearing-impaired community by recognizing specific environmental situations and providing visual alerts. If a model is capable of reliably distinguishing between different sounds and

providing feedback to users without relying on visuals or human input, it opens another dimension of system inputs, ultimately making them much more robust and increasing their capabilities. For these reasons and more, this is not only an interesting problem from a research perspective but also a useful one because of its numerous practical applications.

To train the model, it was important to find a diverse set of already-categorized sound files [6]. Dr. Karol Piczak introduced the set of sound files called ECS-50 with his 2015 paper, "ESC: Dataset for Environmental Sound Classification," which will provide a fantastic starting point for training our model [1]. With 2,000 labeled audio clips, this dataset covers a broad range of sounds, providing comprehensive examples for the model and a strong basis for meaningful performance comparisons. In additional to introducing the audio files, Dr. Piczak also used the cutting-edge methods of his day to show near human accuracy in AI's ability to categorize them.

We will create a baseline through a replication of the original experiment's architecture, as described in Piczak's paper, Environmental Sound Classification with Convolutional Neural Networks [2]. As Piczak mentions in his paper, several challenges were faced at that time, notably, the high computational expense of evaluating a large number of architectural variations. While significant in 2015, those limitations have been reduced through advances in hardware, pretraining, and newer architectures. In the second phase of our experiment, wesd expect to see more accurate results after training the models using current deep learning methods, such as transfer learning from pre-trained models and transformer architectures [3], [4], [5].

Data Source

The primary dataset used in this project is the ESC-50 (Environmental Sound Classification) dataset, originally introduced by Karol J. Piczak in 2015 and made publicly available through Kaggle [6].

The ESC-50 dataset consists of 2,000 labeled audio clips, each 5 seconds in length, organized into 50 distinct classes of environmental sounds, with 40 samples per class. These classes span a wide range of real-world sound categories, including animal sounds (e.g., dog barking), natural sounds (e.g., rain, wind), human activities (e.g., coughing, footsteps), and urban noises (e.g.,

sirens, engine sounds). The dataset has been carefully curated and preprocessed to ensure high-quality labeling and minimal overlap between training and testing samples, making it well-suited for benchmarking machine learning models.

The dataset is distributed as .wav audio files along with accompanying metadata in .csv format, which includes class labels and fold assignments for cross-validation. These predefined folds allow for consistent evaluation and direct comparison with results reported in prior research, including the original study being replicated in this project.

All data required for this project has already been obtained prior to model development. Preprocessing will primarily involve transforming the raw audio signals into log-scaled mel-spectrograms and extracting additional features such as delta coefficients, as described in the original paper. These representations will serve as inputs to the neural network models.

Methods and Tools

The team will employ the methods identified in Table 1 to systematically replicate and extend the foundational architecture introduced by the Piczak model. Initial efforts will focus on reproducing the baseline model configuration, including input feature generation, network topology, and training procedures, to establish a controlled reference for performance benchmarking. This baseline will enable the team to validate implementation correctness while ensuring consistency with reported results in the literature. Once validated, incremental modifications to model structure and training strategy will be introduced to explore potential performance improvements. These modifications will include variations in convolutional layer depth, batch size, learning rate, dropout, and kernel sizes.

Additionally, the team will evaluate model performance using transfer learning by training the ESC-50 dataset on pretrained audio neural networks such as VGGish, YAMNet, and OpenL3. The results of the transfer learning experimentation will be compared to the model performance of the adjusted Piczak model hyperparameters as discussed above.

To rigorously evaluate model performance, the team has defined a comprehensive set of metrics as outlined in Table 2, including classification accuracy, loss convergence behavior, training and inference time, model complexity (measured in number of parameters), and number of epochs required to converge. Comparative analysis will be conducted across all experimental configurations to quantify trade-offs between computational efficiency and predictive performance. This evaluation approach ensures that improvements are not solely driven by accuracy but also considers operational efficiency. Based on these findings, the team will provide targeted recommendations for optimizing model fitness under different constraints.

Table 1: Model Evaluation Methods

CNN Model Layers	Batch Size	Learning Rate	Dropout	Kernel Size
Baseline	Baseline	Baseline	Baseline	Baseline
Baseline	32	1e-3	0.0, 0.1, 0.3, 0.5	3x3, 2x2
3	32	1e-3	0.0, 0.1, 0.3, 0.5	3x3, 2x2
5	32	1e-3	0.0, 0.1, 0.3, 0.5	3x3, 2x2
Baseline	32	1e-2	0.0, 0.1, 0.3, 0.5	3x3, 2x2
3	32	1e-4	0.0, 0.1, 0.3, 0.5	3x3, 2x2
5	32	0.1	0.0, 0.1, 0.3, 0.5	3x3, 2x2
Baseline	100	1e-3	0.0, 0.1, 0.3, 0.5	3x3, 2x2
3	100	1e-3	0.0, 0.1, 0.3, 0.5	3x3, 2x2
5	100	1e-3	0.0, 0.1, 0.3, 0.5	3x3, 2x2

Table 2: Model Results

Convergence Epoch	Training & Validation Accuracy / Loss	Prediction Accuracy / Loss	Training Time / Prediction Time	Model Size (Number of Parameters)
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-

The implementation and experimentation framework will be built using PyTorch and torchaudio, which together provide a flexible and efficient environment for deep learning model development and audio feature extraction. Supporting libraries such as Pandas and Numpy will be used for data preprocessing, dataset management, and numerical operations. Visualization and performance tracking will be conducted using Matplotlib and Seaborn, enabling clear interpretation of training dynamics, validation trends, and prediction outcomes. This will support reproducibility, scalability, and detailed analysis throughout the project development process.

Following initial model evaluation on the baseline dataset, the team will explore data augmentation strategies to further enhance model generalization and robustness. Specifically, manipulating transformations available in torchaudio such as frequency masking, time masking, and time stretching [7]. These augmentation techniques aim to simulate real-world variability in environmental audio signals and reduce overfitting to the training dataset. Through this process, the team seeks to identify optimized preprocessing and training strategies that outperform the

original Piczak approach, ultimately advancing the state of environmental sound classification accuracy and efficiency.

References

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